

There is no unitary operator U acting on two qubits such ...

1. There is no unitary operator U acting on two qubits such that $U|psi\rangle|0\rangle = |psi\rangle|psi\rangle$ for every single-qubit state $|psi\rangle$.
- 1.1. ASSUME: Assume for contradiction that such a unitary U exists: $U|psi\rangle|0\rangle = |psi\rangle|psi\rangle$ for all single-qubit states $|psi\rangle$.
- 1.2. We derive a contradiction by applying U to $|+\rangle = (1/\sqrt{2})(|0\rangle+|1\rangle)$ in two different ways.
- 1.2.1 Applying U to the computational basis states: $U|0\rangle|0\rangle = |0\rangle|0\rangle$ and $U|1\rangle|0\rangle = |1\rangle|1\rangle$.
- 1.2.2 Since U is linear, $U|+\rangle|0\rangle = (1/\sqrt{2})(U|0\rangle|0\rangle + U|1\rangle|0\rangle) = (1/\sqrt{2})(|00\rangle + |11\rangle)$. This is an entangled state.
- 1.2.3 But the cloning assumption gives $U|+\rangle|0\rangle = |+\rangle|+\rangle = (1/2)(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$. This is a product state.
- 1.2.4 Steps 1.2.2 and 1.2.3 give different results for $U|+\rangle|0\rangle$: $(1/\sqrt{2})(|00\rangle+|11\rangle) \neq (1/2)(|00\rangle+|01\rangle+|10\rangle+|11\rangle)$. The left side is entangled (Schmidt rank 2); the right is a product (Schmidt rank 1). Contradiction.